Hellenic Complex Systems Laboratory

The Area Over an ROC Curve as an Index of Diagnostic Inaccuracy

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The Area Over an ROC Curve as an Index of Diagnostic Inaccuracy

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Short Description of the Demonstration

This Demonstration plots the change, the relative change, the rate of change, and the relative rate of change of the area under (blue plot) and the area over (orange plot) the receiver operating characteristic (ROC) curve of a diagnostic test, as the uncertainty of measurement increases from 0 to a user defined upper bound. The test measures a measurand in normally distributed nondiseased and diseased populations, for various values of the means and standard deviations of the populations. A normal distribution of the uncertainty is assumed. The type of plot is selected using the "plot" menu. The five parameters that can be varied using the sliders are measured in arbitrary units.

Search Terms

ROC curves, uncertainty of measurement, sensitivity, specificity, diagnostic test, clinical accuracy, diagnostic accuracy, diagnostic inaccuracy, permissible uncertainty, normal distribution, binormal distribution, area under ROC curves, area over ROC curves, AUC, AOC, relative change, rate of change, relative rate of change

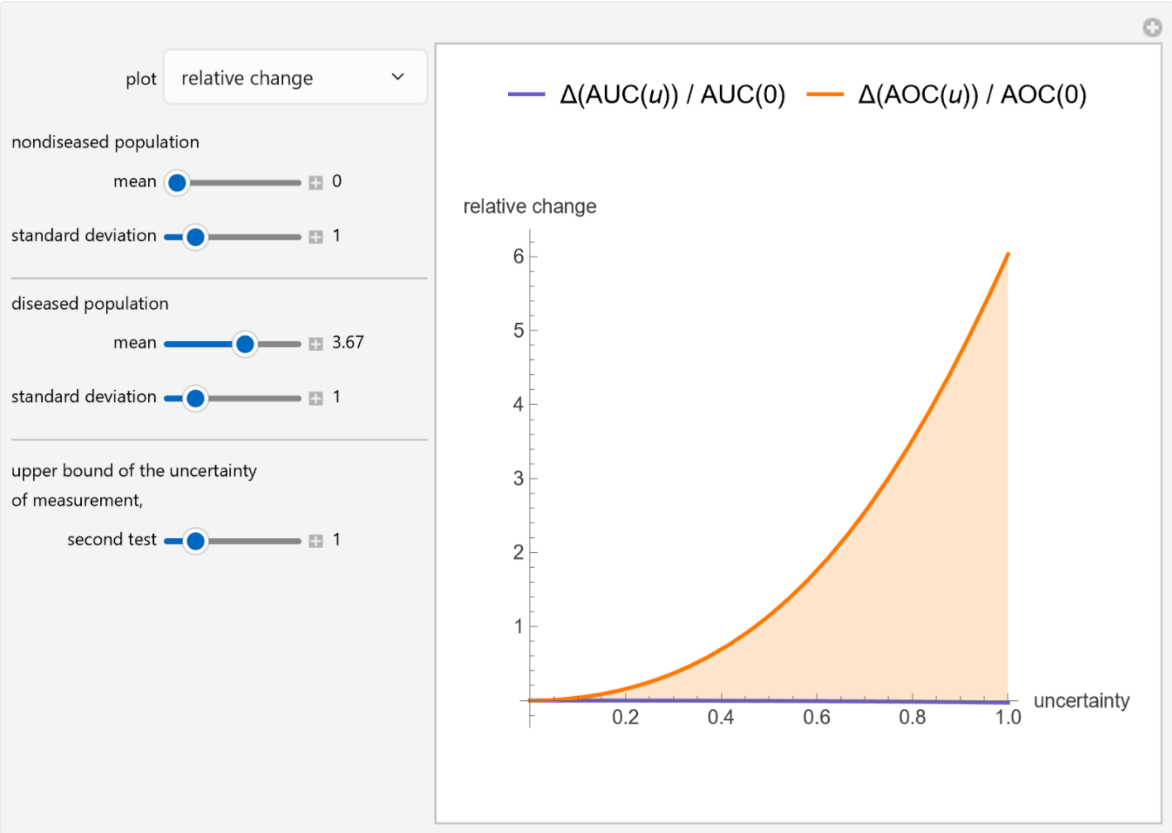

Figure 1: *Plot of the change of the area under (blue plot) and the area over (orange plot) the receiver operating characteristic (ROC) curve of a diagnostic test, with the settings shown at the left.*

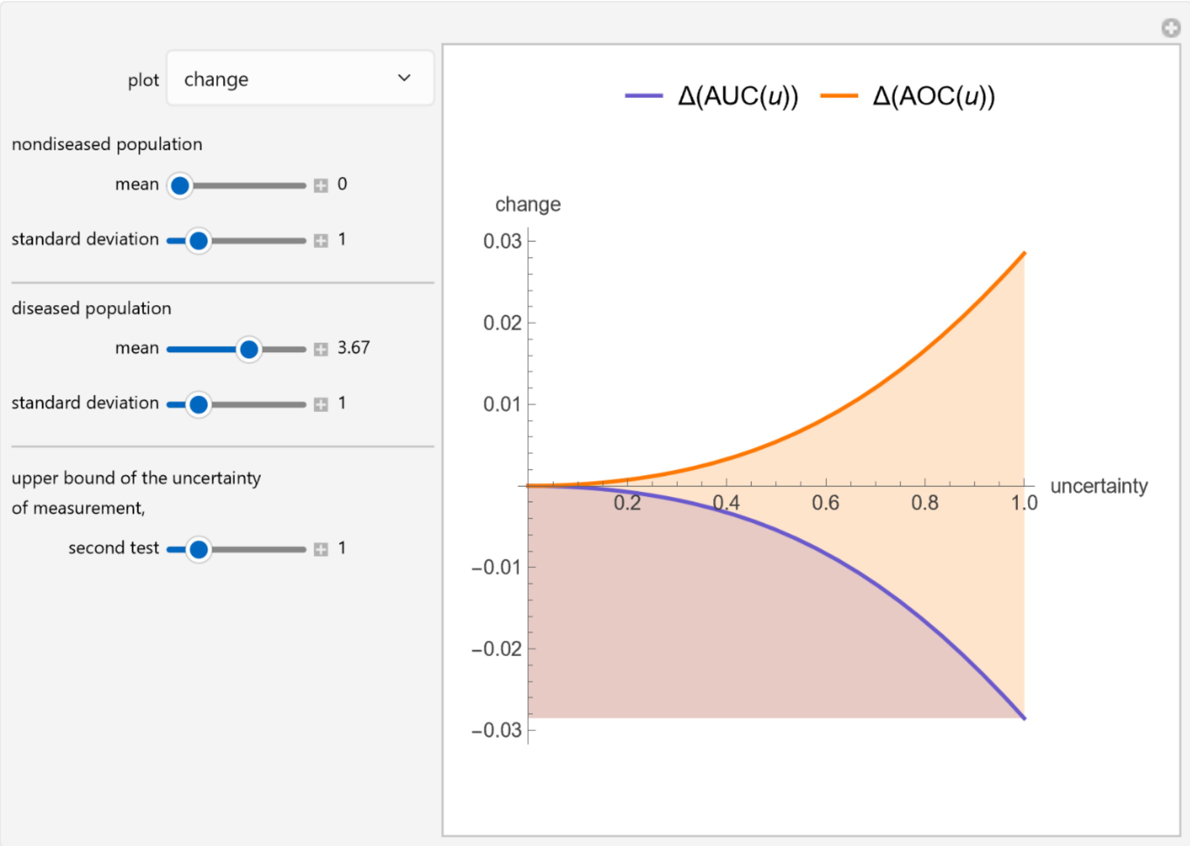

Figure 2: *Plot of the relative change of the area under (blue plot) and the area over (orange plot) the receiver operating characteristic (ROC) curve of a diagnostic test, with the settings shown at the left.*

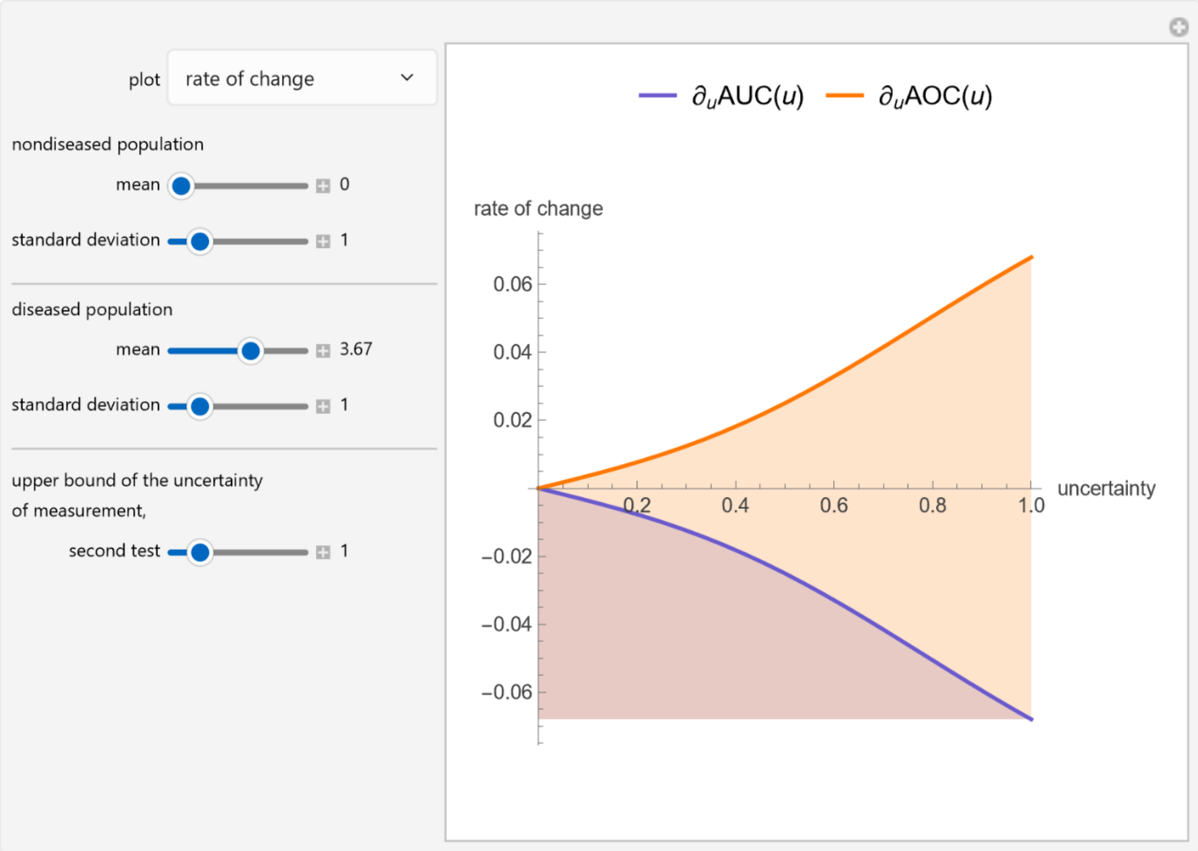

Figure 3: *Plot of the rate of change of the area under (blue plot) and the area over (orange plot) the receiver operating characteristic (ROC) curve of a diagnostic test, with the settings shown at the left.*

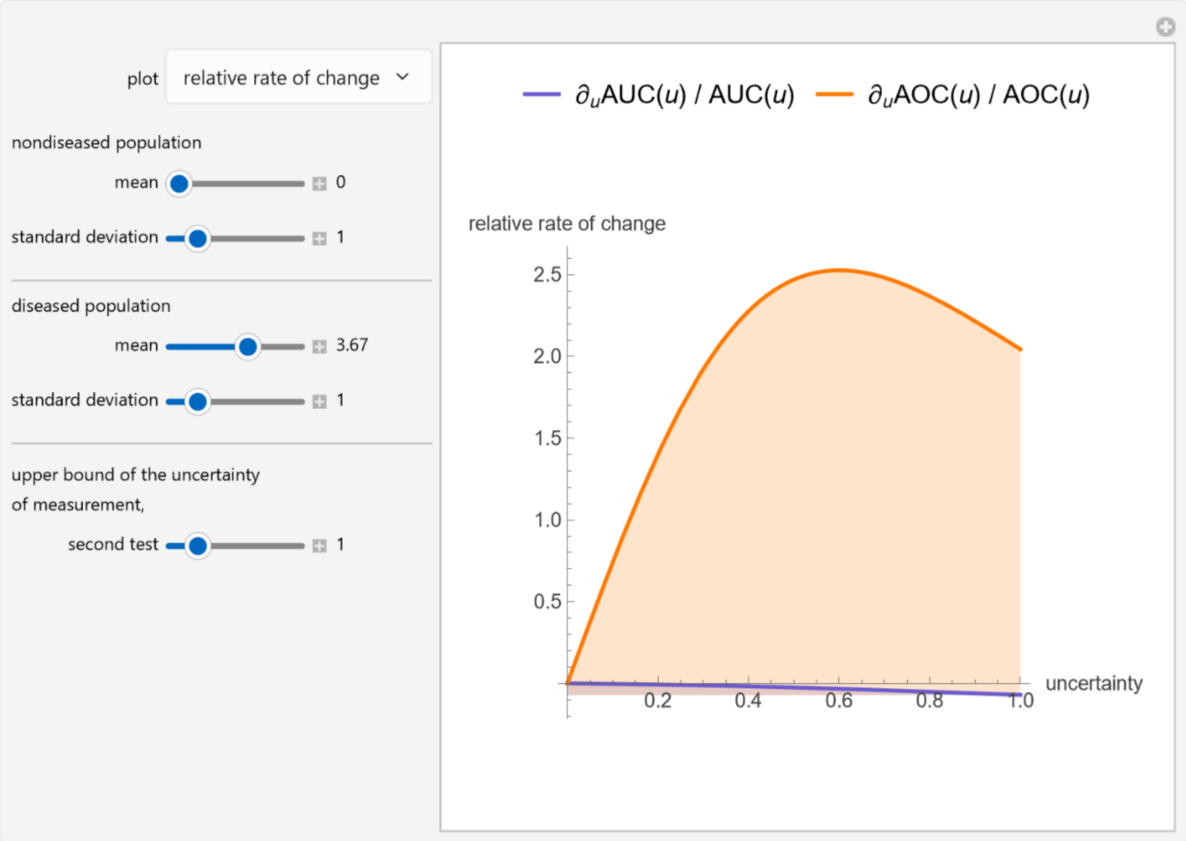

Figure 4: *Plot of the relative rate of change of the area under (blue plot) and the area over (orange plot) the receiver operating characteristic (ROC) curve of a diagnostic test, with the settings shown at the left.*

Details

The ROC curves are used in the evaluation of the clinical accuracy of a diagnostic test applied to a diseased and a nondiseased population. An ROC curve is a plot of the sensitivity of a test against 1-specificity. The term sensitivity is used to describe the fraction of the diseased population with a positive test result, while specificity describes the fraction of the nondiseased population with a negative test result. Therefore, an ROC curve is a plot of the true positive fraction against the false positive fraction. The area under an ROC curve is used as an index of the diagnostic accuracy of the respective test. The area under the curve decreases as the uncertainty of measurement of the diagnostic test increases [1].

Assuming $AUC(u)$ is the area under an ROC curve and $AOC(u)$ is the area over the ROC curve for an uncertainty of measurement u, the changes $\Delta\big(AUC(u)\big)$ and $\Delta\big(AOC(u)\big)$ are defined as $AUC(u) - AUC(0)$ and $AOC(u) - AOC(0)$, the relative changes as $\Delta(AUC(u))/AUC(0)$ and $\Delta(AOC(u))/AOC(0)$, the rates of change as $\frac{\partial AUC(u)}{\partial u}$ and $\frac{\partial AOC(u)}{\partial u}$ and the relative rates of change as $\frac{\frac{\partial AUC(u)}{\partial u}}{AUC(u)}$ and $\frac{\frac{\partial AOC(u)}{\partial u}}{AOC(u)}$ respectively.

As $AOC(u) = 1 - AUC(u)$, it can be considered that the area over the ROC curve is an index of diagnostic inaccuracy. In fact, for $AUC(u) > \frac{1}{2}$, as the plots of this Demonstration show, the relative change and the relative rate of change of the area over an ROC curve against the uncertainty of measurement are greater than the absolute value of the respective measures of the area under the ROC curve, for the same populations.

To the best of my knowledge, measures of the area over the ROC curve against the uncertainty of measurement have not been discussed in the literature, except for [2].

The area over the ROC curve could be used in the evaluation of a diagnostic test, as a diagnostic inaccuracy index. For example, in the thumbnail and the snapshots, the population data describe a bimodal distribution of serum glucose on a non-diabetic and a diabetic population [3].

References:

[1] Hatjimihail AT. *Receiver Operating Characteristic Curves and Uncertainty of Measurement*. Ver 1.1.1. Hellenic Complex Systems Laboratory; 2026. https://doi.org/10.5281/zenodo.20356631

[2] Hatjimihail AT. Uncertainty of Measurement and Areas Over and Under the ROC Curves. Ver. 1.1.1. Hellenic Complex Systems Laboratory; 2026. https://doi.org/10.5281/zenodo.20356939

[3] T. O. Lim, R. Bakri, Z. Morad, and M. A. Hamid, "Bimodality in Blood Glucose Distribution—Is It Universal?" *Diabetes Care,* **25**(12), 2002 pp. 2212-2217.

Demonstration

DOI

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Source

Programming language: Wolfram Language
Availability: The updated source notebook is available at:
https://www.hcsl.com/Tools/Demonstrations/TheAreaOverAReceiverOperatingCharacteristicROCCurveAsAnIndex.nb

First published in 2011 as part of the Wolfram Demonstrations Project:
https://demonstrations.wolfram.com/TheAreaOverAReceiverOperatingCharacteristicROCCurveAsAnIndex/

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS
Other software requirements: Wolfram Player®, freely available at: https://www.wolfram.com/player/ or Wolfram Mathematica®.

System Requirements

Processor: x86-64 compatible CPU.

System memory (RAM): 4 GB+ recommended.

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